Enhancing Sentiment Analysis based Investment by Large Language Models in Japanese Stock Market

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Abstract

This paper explores the utilization of text data as a new α return resource for individual stock investment. In particular, we propose a new construction scheme of a polarity dictionary based on ChatGPT, a state-of-the-art language model, which is also directly used for sentiment analysis. Our research focuses on analyzing non-English text data, expanding the application of sentiment analysis by ChatGPT to a broader range of markets and languages. Through extensive analysis of statistical test, we identify the mean reversion feature following the release of negative news in Japanese large-cap company.

As a result, a simple investment strategy based on the sentiment analysis is superior to the market benchmark index and well-known traditional strategies such as mean-variance and risk parity portfolios.

Keywords: ChatGPT, sentiment analysis, text mining, alpha creation, risk-adjusted returns, natural language processing, Japanese stock market

[*]The findings and conclusions presented in this paper are based on the authors' analysis and interpretation of data. The authors do not guarantee the accuracy or completeness of the information provided. The content of this paper should not be considered as a recommendation or endorsement of any specific investment strategy or security. Investors should exercise their own judgment and seek professional advice before making any investment decisions.

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1 Introduction

It has been well-recognized that stock price prediction is one of the central issues in financial investment. However, it is very challenging because financial markets are driven by the interaction between various factors such as economic, political, and psychological factors [1]. Therefore, it is important to collect and utilize appropriate data which affects the stock price movement as much as possible.

Many researchers have first focused on the market information as shown in [2–5] due to its accessibility. That is, those market information data including return, volatility and dividend yield are mainly represented by numeric values, which are easier to handle with than text data. Note that text data analysis is so difficult that it has become a major research topic as natural language processing (NLP) (e.g., [6–11]).

Nonetheless, text data now attracts more and more attention in financial investment due to the recent development of NLP especially boosted by artificial neural networks (ANNs). Before its appearance, the main stream of sentiment analysis is rule-based approach, where predefined linguistic rules by keywords, patterns, and linguistic heuristics are used to determine sentiment.

For example in financial fields, Tetlock (2007) [12] quantifies the sentiment of the Wall Street Journal column by the fraction of negative words, and finds that news articles with high pessimism increase the downward pressure on the market. Also, Garcia [13] utilizes the proportion of positive and negative words in New York Times to show that the predictability of stock returns by news content is concentrated in recessions. However, these predetermined rule-based systems often lack flexibility and an ability to understand context of the given text data.

Therefore, researchers have developed new approaches such as Word2vec [10] which can interpret the context and meanings of words in a more nuanced manner. Word2Vec, a shallow, two-layered neural network can produce word embeddings, which are vector representations of words learned from large text corpora. These word embeddings can be used as features to capture semantic information and improve sentiment classification performance.

Indeed, this technique has often applied to the financial investment decisions. For instance, Pagolu et al. [14] utilize N-grams and Word2vec to extract features from tweets related to Microsoft stock, which are used as inputs of logistic regression, random forest and SVM models to predict stock price movement. Similarly, Sun et al. [15] apply N-grams and Word2vec to text data collected from microblogs, chat rooms and web forums. As a result, they identify a strong correlation and Granger causality between chat room post sentiments and stock movement in Chinese market.

Furthermore, ANNs have become increasingly important in text data analysis with the development of deep learning technique [16], which realizes fast and precise learning of multi-layered ANNs. As reported in Cybenko [17], the deep-layered complex structure makes it possible to approximate non-linear functions with high accuracy. Consequently, there arises a number of researches which apply ANNs to financial investment problems (e.g., [18–24]). In particular, the recent work by Lopez [24] exploit news headlines to predict stock price movement in U.S. market based on the sentiment score obtained by ChatGPT, which represents a significant advancement in the field of NLP.

Namely, ChatGPT is a large-scale language model which has undergone extensive training on a large corpus of text data such as web pages to grasp the intricacies and patterns of human language based on the Generative Pre-trained Transformer (GPT) architecture. The

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underlying GPT architecture, inspired by the Transformer model introduced in Vaswani et al. (2017) [25], is a deep learning algorithm especially designed for NLP tasks. According to Brown et al. [26], the GPT architecture has consistently achieved state-of-the-art performance across a broad spectrum of NLP tasks including language translation, reading comprehension, and question answering.

Although Lopez [24] shows that ChatGPT excels in both reading comprehension and question answering abilities to predict stock return from text data, it still remains unclear whether these outstanding abilities can be fully demonstrated with the same effectiveness in other languages totally different from English. Also, the short test period, i.e. less than two years, is also a drawback of their work, which makes it difficult to implement statistical analysis.

Hence in order to tackle with these issues, this paper deep dives into the ChatGPT application to the sentiment analysis for stock return prediction. Specifically, we focus on the Japanese language due to the text data accessibility and linguistic difference. Firstly, let us remind that Tokyo Stock Exchange (TSE) in Japan is one of the biggest open market all over the world. Consequently, there exist a vast amount of online accessible news about Japanese stocks, which enables to implement empirical analysis over the long period.

Secondly, the Japanese language is totally different from English. For example, Japanese is an agglutinative language while English is a fusional language. In addition, Japanese consists of three character sets: Hiragana, Katakana, and Kanji, whereas only one character set, i.e. alphabet is used in English.

All these linguistic differences make it difficult for foreigners to enter the Japanese market. In other words, if this research succeeds to demonstrate the usefulness of ChatGPT, the elimination of language barriers may lead to an increase of new investors into the Japanese market, which will be a beneficial shift for both local and global investors. For all these reasons, this paper selects Japanese as the subject of sentiment analysis by ChatGPT.

Here, we briefly review the existing researches about text analysis in Japanese stock market. In Japan, Okimoto and Hirasawa (2014) [27] have pointed out that text information has a strong impact on the TOPIX return in the next day, where positive/negative polarity analysis is determined by news tag information defined in advance. Also, Goshima and Takahashi (2016) [28] uses deep learning techniques to analyze the relationship between Reuters news and TOPIX. Let us remark that these works do not explore the connection between news and individual stocks.

On the other hand, Katayama and Tsuda [29] and Akita [30] show that news information is useful to predict the individual stock price movement. Particularly for sentiment analysis, the former takes a polarity dictionary approach, while the latter adopts Paragraph Vector which extends the idea of Word2Vec to capture the semantic meaning of entire paragraphs or documents.

This paper can be positioned as an extension of these existing studies which examine the relationship between Japanese news and individual stock returns. To the best of our knowledge, this is the first paper which applies ChatGPT to the sentiment analysis on Japanese text data to predict the individual stock returns.

Specifically, we exploit ChatGPT as a non-linear function which outputs sentiment scores from news headlines of individual stocks. For comparison, this paper also calculates sentiment scores based on polarity dictionary. Since the biggest weak point of ChatGPT is its black-box nature of the response generation process, it is important to simultaneously explore interpretable methods such as a word frequency approach.

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As for the construction of the polarity dictionary, we propose a new method using ChatGPT. Namely, ChatGPT is utilized to extract the words which have positive/negative impact on stock price. Let us remark that even if the specific criteria for selecting word sets are not fully transparent, we can understand how the word frequency method categorizes news as positive/negative based on the predefined polarity dictionary.

Moreover, this paper successfully reveals a new feature in Japanese stock market by incorporating company size and time-lag effect of news into the relationship between news headlines and stock returns. Specifically, this paper finds that the large-cap companies belonging to TOPIX Core30 are so robust against bad news that the stock price tends to recover after the first crash just after the news.

As a result, the numerical experiment demonstrates that our approach successfully outperforms the market index, especially for a period from April 2016 to May 2023, during which the benchmark index TOPIX100 records considerable positive return.

The remainder of the paper is organized as follows. Section 2 summarizes text and market return data used in this paper. Section 3 explains our proposed methods for sentiment analysis. Analyzing the relationship between the news and stock return, the performance of the resulting investment strategies are shown in Section 4. After mentioning to the future work in Section 5, Section 6 concludes. Appendix describes details of the performance measures appearing in this paper.

2 Data

Considering liquidity, tradability and quantity of reported news, we focus on 100 companies constituting of TOPIX100 index, though more than one thousand companies are traded on the Tokyo Stock Exchange (TSE). TOPIX100 is a stock price index composed of 100 large-cap stocks with high liquidity among the companies listed on TSE.

In addition, TOPIX100 can be divided into two categories, that is TOPIX Core30 and TOPIX Large70 depending on the market capitalization and liquidity. Compared to TOPIX Large70, TOPIX Core30 is composed of the companies with the larger market capitalization.

Besides, the component stocks are reviewed and replaced once a year in October to ensure that they accurately reflect the current market conditions. 82 and 19 companies remain to be TOPIX100 and TOPIX Core30 during the whole test period, respectively.

2.1 Text Data

We search news headlines related to the 100 companies by using web scraping. For each company, news headlines are collected in the same sample period, i.e. from 2016/04/01 to 2023/05/26. The collected news cover a wide range of topics such as quarterly reports, comments by equity analysts, and public relations. Let us remark that we do not collect the news which directly links to the price movements such as the ones about technical analysis because the aim of this paper is to analyze the impact of the text information apart from the price movements.

As a result, the total number of collected news M is 66,946. Here, we show quantiles of the number of news articles per company in Table 1. This table shows that although there are more than 500 news for each company in average, the number of news articles varies

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significantly across the companies[1]. Also, we note that the correlation between the number of news and market capitalization is 0.65.

Table 1: Descriptive statistics

0%	25%	50%	75%	100%
164	369.75	500.50	750.75	2640

In the following section, we denote the j-th news headline, its distributed timestamp, the whole text dataset and the whole timestamp set as N_j, t_j, WN and TS, respectively. In other words, the whole text dataset WD and the timestamp set TS can be written as $WD = \{N_j\}_{j=1}^{M}$, $\quad TS = \{t_j\}_{j=1}^{M}$, respectively.

2.2 Market Return Data

For numerical experiments, we have downloaded from Bloomberg daily price data $\{P_{t,i}\}_{t,i}$ from 2016/04/01 to 2023/05/26, which are adjusted for all applicable splits and dividend distributions. Let us remark that t and i denote time and company index, respectively. Then, we calculate two types of return data, that is close to close daily return $ON := \{r^{cc}_{t,i}\}_{t,i}$ and open to close intraday return $IN := \{r^{oc}_{t,i}\}_{t,i}$ as follows:

$$r^{cc}_{t,i} := P_{t^c,i}/P_{t-1^c,i} - 1, \qquad r^{oc}_{t,i} := P_{t^c,i}/P_{t^o,i} - 1, \tag{1}$$

where t^o and t^c stand for market opening and closing time[2] on date t, respectively. Table 2 shows the descriptive statistics of the calculated returns, where mean and standard deviation are annualized. For comparison, we also show the close to close daily return of TOPIX100 index r^{tp}_t.

Table 2: Descriptive statistics of the daily and intraday return

	intraday return $\{r^{oc}_{t,i}\}_{t,i}$	daily return $\{r^{cc}_{t,i}\}_{t,i}$	daily TOPIX100 return $\{r^{tp}_t\}_t$
Mean	0.6%	14.3%	11.8%
Standard Deviation	22.4%	30.1%	17.8%
Skew	0.01	0.26	-0.16
Kurtosis	8.64	9.95	7.66

From Table 2, we can see that the mean of intraday return is almost zero, while that of daily return and TOPIX100 return are more than 10%. This result implies that the stock selection and long/short position making are important notably in intraday strategies for individual stocks.

[1]Let us remark that the company with the largest and smallest number of news is Toyota and Japan Exchange Group, respectively.

[2]In Japan, TSE opens at 9:00 and closes at 15:00.

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3 Methodology

3.1 Sentiment Analysis

In this paper, we test two different sentiment analysis. One is word frequency approach based on Bag-of-Words, and the other is ChatGPT approach. The former evaluates news sentiment by counting the number of positive and negative words appearing in the news headline. The latter does by directly asking ChatGPT whether the news has positive/negative impact on the stock price movement. If the calculated score for the j-th news headline S_j^ℓ is greater than/equal to/less than 0, the news is categorized as positive/neutral/negative news, respectively, where $\ell = 0, 1$ denotes the method for calculating the score.

3.1.1 Word frequency approach

This subsection explains how to calculate sentiment score of each news headline by word frequency approach (WFA). First, we implement morphological analysis by using MeCab software[3]. Morphological analysis is a natural language processing technique which divides Japanese text into words and morphemes, i.e. the smallest units of language. Since Japanese sentences lack spaces between words, the decomposition is necessary for word frequency approach based on Bag-of-Words.

Bag-of-Words is a common technique used to represent the features of text data. Particularly, Bag-of-Words has a feature that it treats a document as a collection of words and analyzes based on the frequency or presence of the words without considering the order and context of sentences.

Hereafter, W_j and S_j^0 stands for a set of words obtained by MeCab and a sentiment score for the j-th news headline, respectively. Also, let D_p and D_n be a set of positive and negative words, respectively. Then, we define S_j^0 as follows:

$$S_j^0 := \#\{w \in W_j \cap D_p\} - \#\{w \in W_j \cap D_n\}, \tag{2}$$

where $\#A$ is its counting measure, i.e. $\#A := \sum_{w \in A} 1$.

From this definition, we can easily know that the components of D_p and D_n, also known as a polarity dictionary, has a great impact on the sentiment score S_j^0. While the previous researches construct those polarity dictionaries based on human judgment, this paper utilizes ChatGPT for its construction as described below.

First, we construct a new set FW whose components are the top 1,000 words frequently appearing in the whole news headlines. Concretely, we construct a set $\mathrm{WNH}_1 := \cup_{j=1}^M W_j$ and a multiset $\mathrm{WNH}_2 := \sum_{j=1}^M W_j$ from a set of words W_j. Let us remark that a set WNH_1 is not equal to a multiset WNH_2 because a multiset allows for multiple instances for each of its element. Then, FW is defined as follows:

$$\begin{aligned} FW &:= \{w \in \mathrm{WNH}_1 \mid \#\{w \in \mathrm{WNH}_2\} \geq A[1000]\}, \\ A &:= \{\{\#\{w \in \mathrm{WNH}_2\} \mid \forall w \in \mathrm{WNH}_1\}\}, \end{aligned} \tag{3}$$

where $\{\{\cdot\}\}$ and $A[n]$ denotes a multiset and the n-th largest element in set A, respectively.

Then, we ask ChatGPT as follows:

[3]See Kudo (2005) [31] for the description of the software.

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Extract the words which have positive/negative impact on stock price movement in general from the below word set FW.
$FW = \{[FW_1],\ [FW_2],\ \cdots,\ [FW_{1000}]\}$

Let us note that ChatGPT can not process a word set WNH_1 since WNH_1 is an extremely large word set. Therefore, we prepare an extracted word set $FW \subset \mathrm{WNH}_1$. The answers from ChatGPT for positive/negative are used as the elements of D_p/D_n. As a result, 25 and 16 words are selected as D_p and D_n, respectively, which are listed in below Table 3 and 4.

Table 3: D_p: positive words selected by ChatGPT

注目 (attention)	買い (buy)	好感 (positive sentiment)	買収 (acquisition)
好調 (strong performance)	支援 (support)	好 (positive)	高 (high)
反発 (rebound)	上方 (upward)	高い (high)	増益 (increase in profit)
増額 (increase)	高まる (rise)	上昇 (rise)	続伸 (surge)
期待 (expectation)	成長 (growth)	収益 (revenue)	合意 (agreement)
活用 (utilization)	主導 (lead)	導入 (introduction)	参入 (entry)
増資 (capital increase)			

Table 4: D_n: negative words selected by ChatGPT

下落 (decline)	続落 (continued decline)	売り (sell)	売却 (sale)
値下がり (price decline)	下値 (lower price)	急落 (plunge)	減益 (decrease in profit)
下方 (downward)	逆風 (headwind)	マイナス (negative)	空売り (short sale)
売り出し (offering)	売ら (sold)	下回る (below)	ネガティブ (negative)

From these tables, we can see that the selected words are consistent with human sense and intuition. Using these polarity dictionaries, we calculate the sentiment score S_j^0.

3.1.2 ChatGPT approach

In this subsection, we explain how to calculate sentiment score by ChatGPT, which is a large-scale language model developed by OpenAI based on GPT architecture. Specifically, we exploit ChatGPT[4] as a non-linear function $cgpt : WD \to \{-1, 0, 1\}$ by asking ChatGPT as follows:

> Forget all your previous instructions. Pretend you are a financial expert with stock recommendation experience. Is this headline good or bad for the stock price of Company i_j?
> Headline: "N_j"
> Answer 1 if good news, -1 if bad news, or 0 if uncertain in the first line.

where Company i_j means the company appearing in the j-th news N_j. In other words, we obtain sentiment score S_j^1 by $S_j^1 := cgpt(N_j)$.

As clearly shown by this definition, ChatGPT inputs the raw text data N_j, which implies that it considers the order and context of sentences while the word frequency approach does not. In addition, the word frequency approach cannot analyze the sentiment of text data

[4]We use gpt-3.5-turbo as of June 8.

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which do not include the sentiment word as explained in the previous section. Hence, the results by ChatGPT is expected to be different from those by the word frequency approach.

4 Empirical Study

After obtaining the sentiment scores S_j^0 and S_j^1, we measure their effectiveness as an α-return driver. In order to evaluate the effectiveness of the sentiment scores as an α-return driver, we implement two numerical experiments: One is a statistical test and the other is investment simulation.

In the following numerical experiments, we assume that a position can be taken only at the market opening and closing time considering the liquidity. Although news information has the greatest impact at its reported timing, the impact is extremely large enough to deplete the liquidity rapidly. As a result, only limited investors such as algo traders can take the price at the moment when the news is released. In other words, it is unrealistic to assume that retail investors and institutional investors with large investment sizes can trade securities with the price just at the moment.

Concretely, we take a position at the coming market closing time if news is reported during trading hours. Else, we take a position at the closest opening time. Also, the position is closed at the next market closing time after taking a position.

4.1 Statistical test analysis

In this section, we first implement statistical test to confirm that the mean of the stock return after the news is different from that without news. Let us remind that skew and kurtosis in Table 2 implies that the return of individual stock does not follow normal distribution, which is confirmed by Shapiro-Wilk test [32]. Hence, Wilcoxon-Mann-Whitney (WMW) test [33] is used in the following analysis.

In particular, we construct 8 return data sets $RD_{0,k}^{\ell}$ and $\overline{RD}_{0,k}^{\ell}$ ($k = 0, 1$, $\ell = 0, 1$), which are a dataset of the daily and intraday return just after the news distribution, respectively. Concretely, they are defined as follows:

$$
\begin{aligned}
RD_{0,0}^{\ell} &:= \{r_{t+1,i_j}^{cc} \in ON \mid j \in \{1, \cdots, M\},\ t_j \in [t^o, t^c),\ S_j^{\ell} > 0\}, \quad \ell = 0, 1, \\
RD_{0,1}^{\ell} &:= \{r_{t+1,i_j}^{cc} \in ON \mid j \in \{1, \cdots, M\},\ t_j \in [t^o, t^c),\ S_j^{\ell} < 0\}, \quad \ell = 0, 1, \\
\overline{RD}_{0,0}^{\ell} &:= \{r_{t+1,i_j}^{oc} \in IN \mid j \in \{1, \cdots, M\},\ t_j \in [t^c, t+1^o),\ S_j^{\ell} > 0\}, \quad \ell = 0, 1, \\
\overline{RD}_{0,1}^{\ell} &:= \{r_{t+1,i_j}^{oc} \in IN \mid j \in \{1, \cdots, M\},\ t_j \in [t^c, t+1^o),\ S_j^{\ell} < 0\}, \quad \ell = 0, 1,
\end{aligned}
\tag{4}
$$

where index k and ℓ means the news sentiment and the method of obtaining the sentiment score, respectively. Then, $RD_{0,k}^{\ell}$ and $\overline{RD}_{0,k}^{\ell}$ are compared with $ON_{0,k}^{\ell} := ON \backslash (RD_{0,0}^{\ell} \cup RD_{0,1}^{\ell})$ and $IN_{0,k}^{\ell} := IN \setminus (\overline{RD}_{0,0}^{\ell} \cup \overline{RD}_{0,1}^{\ell})$ to reject the null hypothesis that the mean of return set after the positive/negative news is less/greater than usual.

Table 5: Comparison of statistical test results between WFA and ChatGPT

	$RD_{0,0}^0$	$RD_{0,1}^0$	$\overline{RD}_{0,0}^0$	$\overline{RD}_{0,1}^0$	$RD_{0,0}^1$	$RD_{0,1}^1$	$\overline{RD}_{0,0}^1$	$\overline{RD}_{0,1}^1$
p-value	**0.0%**	33.3%	7.9%	6.3%	**0.0%**	**2.9%**	**1.2%**	**3.2%**
Mean	32.9%	-6.8%	1.2%	-19.9%	30.2%	-16.3%	4.6%	-20.9%

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Table5 summarizes the p-value of WMW test and an annualized mean of each set. This table shows that ChatGPT approach can reject null hypothesis in all cases at 5.0% significance level, while traditional word frequency approach can only in one case. Moreover, by comparing this table with Table2, we can see that the average return after the positive/negative news is much larger/smaller than usual except $\overline{RD}^0_{0,0}$, respectively. Overall, these results indicate that news information still influences stock price movements even after the first market impact at the reported moment.

4.2 Difference between WFA and ChatGPT

In this section, we compare the categorization difference between word frequency approach and ChatGPT. Table 6 summarizes the number of news categorized as positive/neutral/negative by each method.

Table 6: Number of news articles classified by sentiment analysis

		ChatGPT			
		positive	neutral	negative	total
WFA	positive	17,155	9,201	768	27,124
	neutral	8,135	24,391	3,208	35,734
	negative	187	264	3,637	4,088
	total	25,477	33,856	7,613	66,946

This table shows that the sentiment scores calculated by the two approaches are generally consistent with each other. That is, 67.5% (=45,171/66,924) of all news are categorized into the same class by both methods, while only 1.4% (=955/66,924) have totally opposite classification.

Now for the differences between ChatGPT and WFA, we examine the news articles whose scores differ depending on the calculation method. Specifically, the following news[5] are regarded as positive by ChatGPT, while neutral/negative by WFA, respectively.

- Recruit HD continues to surge significantly, leading to a bull market. Recent strong performance resists downward pressure.
- AEON is performing well, with the report on the brisk New Year's first sales.

Since the former news has one positive word "surge" and one negative word "downward", WFA judges it as neutral. Also, the latter news has one negative word "sale", it is regarded as negative by WFA. However considering the context, both news are apparently positive for Recruit HD and AEON, respectively, which is appropriately captured by ChatGPT. These examples clearly show the limitation of WFA and the advantage of ChatGPT that it can properly interpret the context of text data written in Japanese.

Furthermore in order to calculate the hit ratios, we breakdown Table 6 into Table 7 depending on the sign of the return $RD^{\ell}_{0,k}$ and $\overline{RD}^{\ell}_{0,k}$, where the value of the left (right) side indicates the number of news articles with positive (negative) return on the next day.

[5]Although the raw text data is written in Japanese, it is translated into English for non-Japanese readers.

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Table 7: Breakdown of the number of news depending on the return on the next day

		ChatGPT			
		positive	neutral	negative	total
WFA	positive	8,717 : 8,438	4,591 : 4,610	364 : 404	13,672 : 13,452
	neutral	4,075 : 4,060	12,202 : 12,189	1,607 : 1,601	17,884 : 17,850
	negative	95 : 92	125 : 139	1,809 : 1,828	2,029 : 2,059
	total	12,887 : 12,590	16,918 : 16,938	3780 : 3833	33,585 : 33,361

Combined Table 7 with Table 6, we can calculate the hit ratios. That is, the hit ratios for the positive news by ChatGPT and WFA are 50.6% (=12,887/25,477) and 50.4% (=13,672/27,124), while 50.3% (=3,833/7,613) and 50.4% (=2,059/4,088) for the negative news, respectively.

Considering the results of the mean in Table 5, these mediocre hit ratios suggest that both approaches have a strong advantage in predicting significant positive and negative returns rather than improving overall accuracy of hit ratios. Here, let us note that when ChatGPT and WFA categorize the same news into a totally opposite class, ChatGPT tends to provide more accurate responses, though there are only little differences in terms of hit ratios.

4.3 Difference of news sensitivity by company size

This section investigates the difference of news sensitivity depending on the company size. Specifically, the component companies of TOPIX100 are divided into two groups: Core30 and Large70. With regard to each group, we implement a similar analysis as in Section 4.1. Let us remark that we only present the results based on ChatGPT method in the following analysis for simplicity.

Table 8: Comparison of statistical test results between Core30 and Large70

	Core 30				Large 70			
	$RD^1_{0,0}$	$RD^1_{0,1}$	$\overline{RD}^1_{0,0}$	$\overline{RD}^1_{0,1}$	$RD^1_{0,0}$	$RD^1_{0,1}$	$\overline{RD}^1_{0,0}$	$\overline{RD}^1_{0,1}$
p-value	**0.0%**	**4.4%**	49.4%	63.1%	**2.7%**	17.1%	**0.2%**	**0.8%**
Mean	31.9%	-16.0%	-2.3%	2.8%	29.0%	-16.5%	8.8%	-31.4%
SD	1.86%	2.08%	1.47%	1.94%	2.22%	2.45%	1.63%	2.33%

Table 8 summarizes the p-value of WMW test and an annualized mean of each set in addition to the standard deviation (SD). As for the news during trading hours $RD^{\ell}_{0,k}$, the mean of the return after the positive/negative news is much larger/smaller than market average in both cases, respectively, though Large70 after the negative news $RD^1_{0,1}$ fails to reject the null hypothesis due to the larger standard deviation.

With regard to the news before and after trading hours $\overline{RD}^{\ell}_{0,k}$, the results are largely different from the previous ones especially in Core30. Concretely, there is a consistent positive relationship between the sentiment score and the returns for the Large70 stocks, which is confirmed to be statistically significant. On the other hand, the relationship is totally reversed in the case of Core30 stocks. That is, the intraday return after the positive news is -2.3% in average, while +2.8% after the negative news.

This reversal phenomenon is attributed to the rapid overreaction at the market opening time. Since Core30 companies attract more attention from investors, the market tends to

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react more quickly and superfluously to the news, which makes the opening price overvalued. As time passes, the market tends to cool down with the reversion to an appropriate levels at the closing time. Since sentiment analysis of news headlines alone cannot respond to the market real-time dynamics, it is important to combine our proposed method with other data sources such as price data in order to effectively adapt the market dynamic environment.

4.4 Delayed News Effect

In this section, we explore the time-lag effect of the news. Although we only focus on the returns of the next day following the news announcement in the previous sections, there is a possibility that the impact of the news may appear with delay. Hence, we research the delayed news effect by constructing the n-days after daily return set $RD^{\ell}_{n,0}$ and $RD^{\ell}_{n,1}$.

$$\begin{aligned} RD^{\ell}_{n,0} &:= \{r^{cc}_{t+1+n,i_j} \in ON \mid t_j \in [t^o, t+1^o) \wedge \ S^{\ell}_j > 0, \ \forall j \in \{1, \cdots, M\}\}, \quad \ell = 1, \ n = 1, \cdots, 10, \\ RD^{\ell}_{n,1} &:= \{r^{cc}_{t+1+n,i_j} \in ON \mid t_j \in [t^o, t+1^o) \wedge \ S^{\ell}_j < 0, \ \forall j \in \{1, \cdots, M\}\}, \quad \ell = 1, \ n = 1, \cdots, 10. \end{aligned} \tag{5}$$

Note that we focus on the ChatGPT case $\ell = 1$ for simplicity again as in the previous section. Also, we set the maximum value of n as 10 to investigate the impact up to two weeks later.

In the same way as the previous sections, we apply a WMW-test to this n-day after daily return set to reject the null hypothesis that the mean of the return set is equal to that without news. Table 9 summarizes the mean of each return set where the bold texts indicate that the null hypothesis is rejected at the two-sided 5.0% significance level.

Table 9: Mean of n-day after return set: All, Core30 and Large70

	All		Core30		Large70	
n	positive	negative	positive	negative	positive	negative
1	**19.6%**	8.4%	**22.2%**	6.9%	**17.9%**	9.5%
2	12.1%	27.2%	15.3%	**36.8%**	10.0%	20.1%
3	6.7%	11.5%	3.3%	10.0%	8.9%	12.7%
4	10.7%	23.4%	9.1%	**34.1%**	11.7%	15.6%
5	5.5%	3.3%	2.3%	7.6%	7.6%	0.2%
6	11.3%	28.1%	13.2%	38.1%	10.0%	20.9%
7	**17.8%**	21.3%	18.6%	**27.4%**	17.3%	16.8%
8	14.7%	21.1%	20.2%	12.1%	11.1%	**27.7%**
9	10.5%	16.3%	7.9%	30.0%	12.2%	6.2%
10	10.4%	8.0%	8.8%	10.3%	11.4%	6.3%

First of all, we can see that the effect of positive news persists for an additional day in both Core30 and Large70, though the value of the mean is smaller than the case $n = 0$. On the other hand, the downtrend by negative news does not last, which implies that the market incorporates negative news more rapidly.

Moreover with regard to Core30 companies, the stock price tends to recover a few days after the negative news. Namely, the results in Table 9 confirms that the means over 30% are significantly different from usual times in the case of $n = 2$ and 4. Considering that this recovery feature is unique to Core30, the reason of this phenomenon can be attributed to the robustness of Core30 companies.

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Since the companies selected for Core30 are typically large and stable, investors buy back the stock with relief a few days after the negative news based on the idea that the negative factors have been fully reflected in the stock price, which causes the price recovery. This finding is interesting and valuable for investors because it suggests potential opportunities for contrarian trading strategies based on sentiment analysis, which are closely investigated in the next section.

4.5 Investment performance

This subsection presents the results for investment performances, namely, comparison of sentiment analysis based strategy with market benchmark index, i.e. TOPIX100 and traditional strategies including mean-variance (MV) and risk-parity (RP) portfolios[6].

Concretely, we construct the following four types of portfolio:

(i) Sentiment Trend Following Equally Weighted Portfolio (STFEW):
Take a long/short position on the next day following the news announcement, i.e. in the case of $n = 0$. The position side (long/short) is decided depending on the sign of the sentiment score.

(ii) Adjusted-Sentiment Trend Following Equally Weighted Portfolio (ASTFEW):
Take a position in the same way as (i) STFEW. However considering the results in Section 4.3, we do not take an intraday position for Core30 companies.

(iii) Sentiment Contrarian Equally Weighted Portfolio (SCEW):
Considering the results in Section4.4, take a long positions during 2 to 4-day after the negative news announcement, i.e. in the case of $n = 2, 3, 4$.

(iv) Sentiment Trend Following and Contrarian Equally Weighted Portfolio (STFCEW):
Take a position according to both STFEW and SCEW.

Although the long/short position side differs depending on the sentiment score, all strategies allocate equally weighted position to each stock. That is, the absolute value of each position is $1/N_t$ where N_t denotes the number of stocks investing at time t. Let us remark that we average the sentiment scores if there are multiple news headlines for a company during a given trading or non-trading hours.

4.5.1 STFEW: basic strategy based on sentiment analysis

First, we focus on the investment performance of the most basic strategy, i.e. STFEW, which are shown in Table 10[7] and Fig. 1. Let us note that all the figures in this paper is a logarithmic graph with a base of 2.

[6]For the construction of these traditional portfolios, it is necessary to estimate the expected return and covariance matrix of the stocks. This paper adopts 3-months exponential moving average for the estimation.

[7]See appendix for the details of the performance measures.

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Table 10: Performance comparison: STFEW v.s. traditional portfolio

	MV	RP	TOPIX100	STFEW by WFA	STFEW by ChatGPT
CR	2.2%	10.1%	10.1%	14.4%	16.8%
SD	15.6%	15.6%	17.2%	13.5%	11.5%
DD	10.9%	11.0%	12.1%	9.0%	7.5%
MDD	41.0%	28.0%	28.7%	18.1%	14.7%
ShR	22.0%	69.4%	64.3%	106.6%	140.9%
SoR	31.3%	98.5%	91.8%	159.9%	215.7%
StR	8.3%	38.5%	38.7%	79.5%	110.4%

Fig. 1: Performance comparison: STFEW v.s. traditional portfolio

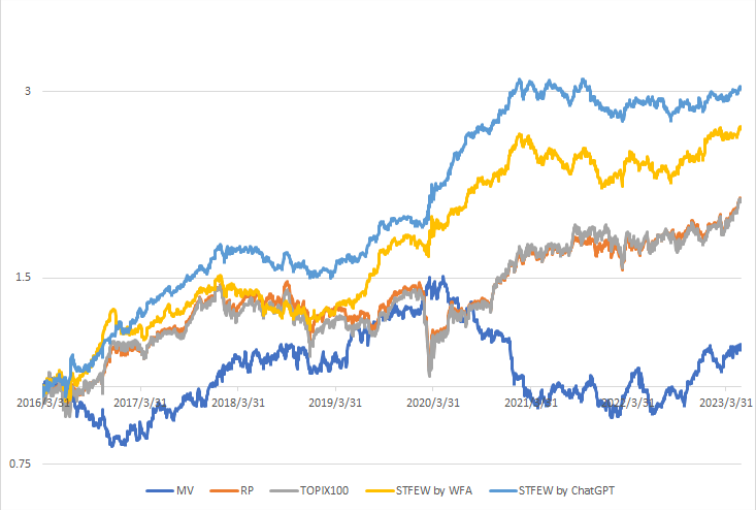


Table 10 summarizes the investment performance of STFEW based on both WFA and ChatGPT as well as the traditional portfolios and TOPIX100. From this table, we can see that our sentiment analysis based scheme remarkably outperforms the traditional strategies as well as TOPIX100 in all performance measures.

Specifically, CR and risk-adjusted returns, i.e. ShR, SoR and StR are higher, while risk measures, i.e. SD, DD and MDD are lower than the traditional ones, which are also confirmed by Fig. 1 demonstrating the performance transition of each strategy. This figure clearly shows that STFEW does not suffer serious drawdown during the Covid shock period (2020/2/06-2020/3/16).

In terms of the comparison between WFA and ChatGPT, the performance of ChatGPT is superior to that of WFA, which is consistent to Section 4.1. This result again suggests that the ability of ChatGPT to read and understand context is also crucial in enhancing investment performance.

4.5.2 The enhanced strategies based on ChatGPT

Next, let us confirm the results of the enhanced strategies (ii)-(iv) by incorporating the market features discovered in Section 4.3 and 4.4. Here, we do not present the results based on WFA

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for readability, which are shown in the next subsection.

Table 11: Performance comparison: the enhanced strategies based on sentiment analysis

	MV	RP	TOPIX100	STFEW	ASTFEW	SCEW	STFCEW
CR	2.2%	10.1%	10.1%	16.8%	29.2%	18.3%	17.5%
SD	15.6%	15.6%	17.2%	11.5%	17.2%	22.4%	11.5%
DD	10.9%	11.0%	12.1%	7.5%	11.1%	14.8%	7.6%
MDD	41.0%	28.0%	28.7%	14.7%	22.0%	35.0%	13.6%
ShR	22.0%	69.4%	64.3%	140.9%	157.4%	86.3%	146.6%
SoR	31.3%	98.5%	91.8%	215.7%	244.5%	130.5%	222.2%
StR	8.3%	38.5%	38.7%	110.4%	123.3%	55.1%	123.4%

Fig. 2: Performance comparison: the enhanced strategies based on sentiment analysis

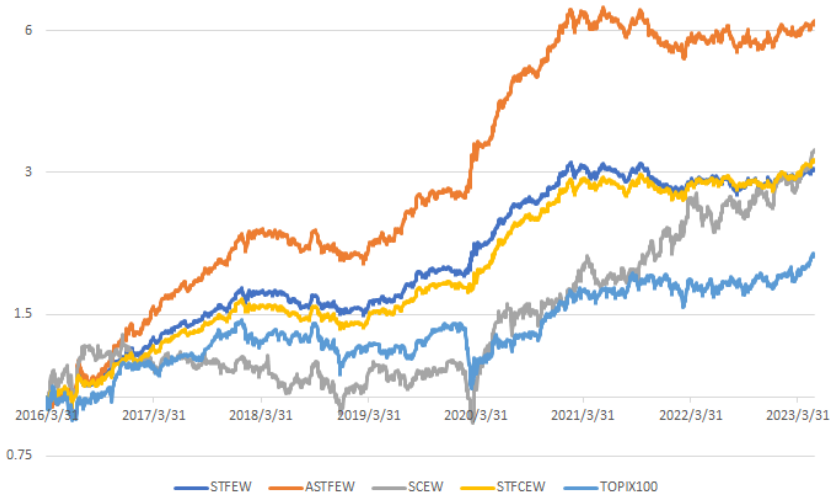


Table 11 and Fig. 2 shows the investment performances of the enhanced strategies. From Table 11, it is clear that all of the enhanced strategies again outperforms the traditional portfolios and TOPIX100 in terms of risk-adjusted returns. In particular, the performance of ASTFEW stands out remarkably well, which implies the significant impact of the overshooting effect observed in Core30.

As for a contrarian strategy SCEW, although the performance is relatively low compared to STFEW and ATFEW, it attains the low correlation 0.29 to the STFEW with the better performances than TOPIX100. Specifically, both STFEW and ATFEW cannot update the high-water mark[8] for the last two years, while SCEW is almost doubled during that period. Therefore, STFCEW, the combination of STFEW and SCEW, improves the performance of STFEW, which records the highest StR in all strategies.

Overall, the results of the numerical experiment conclude that our proposed portfolios based on sentiment analysis are effective to improve the performance of Japanese stock in-

[8]Here, a high-water mark means the highest peak in value that an investment strategy has reached.

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vestment.

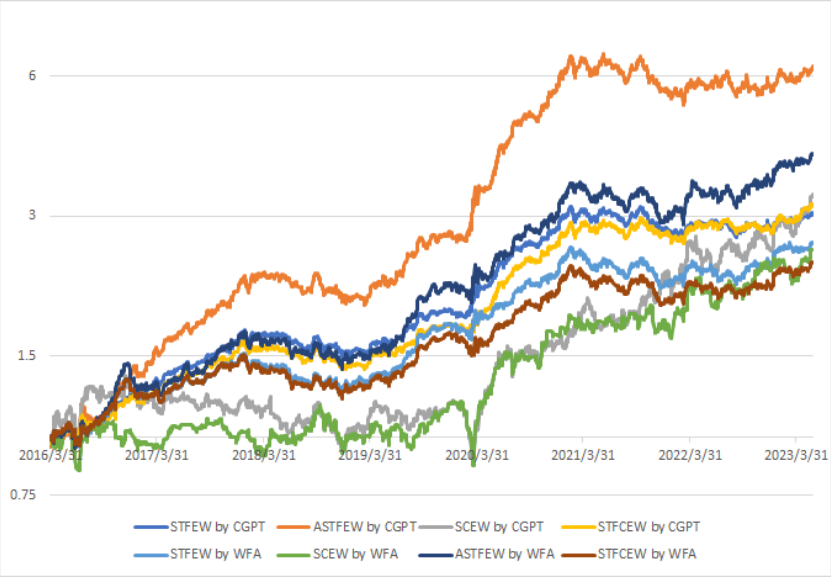
4.5.3 The enhanced strategies based on WFA

Finally, we show the results of the enhanced strategies based on WFA by Fig. 3 and Table 12, where we also presents the correlation to the corresponding ChatGPT based strategy (CGPT Cor).

Table 12: Performance comparison: ASTFEW, SCEW and STFCEW based on WFA

	MV	RP	TOPIX100	STFEW	ASTFEW	SCEW	STFCEW
CR	2.2%	10.1%	10.1%	14.4%	21.7%	13.8%	12.9%
SD	15.6%	15.6%	17.2%	13.5%	17.2%	21.0%	13.3%
DD	10.9%	11.0%	12.1%	9.0%	11.1%	13.5%	9.1%
MDD	41.0%	28.0%	28.7%	18.1%	19.1%	27.1%	19.7%
ShR	22.0%	69.4%	64.3%	106.6%	122.4%	72.0%	97.6%
SoR	31.3%	98.5%	91.8%	159.9%	189.5%	112.0%	143.6%
StR	8.3%	38.5%	38.7%	79.5%	110.8%	55.9%	66.0%
CGPT Cor	-	-	-	75.7%	72.1%	70.2%	80.2%

Fig. 3: Performance comparison: ASTFEW, SCEW and STFCEW based on WFA



Although the performance based on WFA is worse than ChatGPT, it can attain the higher performance than that of MV, RP and TOPIX100. Interestingly, the correlation between ChatGPT and WFA is not particularly high considering the fact that the correlation between STFEW and ANTFEW is over 95%.

Specifically from Fig. 3, we can see that ASTFEW based on WFA can update the high-water mark in the last two years, while that based on ChatGPT cannnot. Let us note that the previous research [5] with price data reports that the model performance can be improved by

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diversifying the estimation methods for expected returns to take a model averaging approach. Likewise, diversification for the method of sentiment analysis may improve the investment performance, which is also one of our future research topics.

5 Future work

For the further development of sentiment analysis using ChatGPT, we closely confirm the case in which the stock price on the next day moves toward the opposite direction to the one suggested by sentiment analysis based on ChatGPT. Specifically, the following news, released after the trading hour on 2019/08/05, is judged as negative both by ChatGPT and WFA.

- Suzuki reported a 46% decrease in ordinary income for the April-June 1st quarter.

Actually, the stock price at the market open declines by 9.0% from the closing price on the previous day, which supports the validity of the sentiment analysis. However, the stock price increases by 10% from the opening price at the market closing time, which causes serious damages to our proposed investment strategy.

As stated in Section4.3, the market often instantly overreacts to the news, followed by an adjustment period during which the price moves toward the opposite direction. Therefore in order to improve the investment performance, it is also important to find out whether the market is currently overheating or not, which is among the research areas we plan to explore in the future.

Also, further research can focus on exploring and developing additional language models other than ChatGPT. Although the hit ratios by our proposed method are almost 50% as shown in Section4.2, there is a room for improvement by the use of the better language models, which is also one of our future research topics.

Moreover, all of the strategies proposed in this paper are an extension of equally-weighted strategy which do not use the traditional price information. Since there are a number of researches for financial investment based on the price information, the combination of the existing approach with the sentiment analysis seems to further boost the investment performance. For instance, it is interesting to develop a new time series model with a sentiment score term as an explanatory variable.

6 Conclusion

In this paper, we have investigated a novel approach of sentiment analysis for stock investment based on ChatGPT and introduced a new construction scheme for a polarity dictionary. Our research has extended the scope of sentiment analysis to include non-English Japanese text data, allowing for broader market coverage and enhanced cross-lingual analysis.

To show the validity of our scheme, we have implemented a WMH test, which clarifies that the features extracted from news headlines through ChatGPT-based sentiment analysis have statistically significant effect on investment performance. Moreover, through our empirical analysis, we have made significant contributions to the understanding of sentiment-driven market dynamics. Specifically, we have identified the mean reversion characteristics following negative news for large-cap stocks.

This finding offers valuable insights for investors, suggesting potential opportunities for contrarian trading strategies based on sentiment analysis, which is confirmed in our numerical

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simulation. In particular, our numerical results have shown that the simple investment strategy based on the sentiment analysis outperforms the benchmark market index and well-known traditional strategies such as mean-variance and risk parity portfolios.

Although this paper utilizes ChatGPT as it is, the hit ratios achieved by our approaches are approximately 50%. Hence, it is also important to explore and develop an alternative language models beyond ChatGPT, which is one of our future research topics.

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We really appreciate Kyo Yamamoto and Soichiro Takahashi at GCI Asset Management Inc. for their valuable comments.

Appendix: Performance measure

Here, we summarize the definition of performance measures used in the current paper.

- Compound return (CR):

$$CR \equiv \left\{ \prod_{t=1}^{T} (1 + R_t) \right\}^{1/T} - 1. \tag{6}$$

  This is one of the most fundamental performance measures, which corresponds with a geometric average of the portfolio returns $\{R_t\}$.

- Standard deviation (SD):

$$SD \equiv \left\{ \frac{1}{T} \sum_{t=1}^{T} (R_t - \bar{R})^2 \right\}^{1/2}, \ \bar{R} \equiv \frac{1}{T} \sum_{t=1}^{T} R_t. \tag{7}$$

  This is one of the most basic variables both in theory and practice for portfolio risk management or derivatives pricing and hedging, which is also known as volatility.

- Downside deviation (DD):

$$DD \equiv \left\{ \frac{1}{T} \sum_{t=1}^{T} \min(0, R_t)^2 \right\}^{1/2}. \tag{8}$$

  Differently from SD, this risk measure regards only negative return as risk, which seems reasonable for investment performance evaluation.

- Maximum drawdown (MDD):

$$MDD \equiv \max_{1 \le t \le T} \frac{M_t - V_t}{M_t}, \quad M_t \equiv \max_{0 \le s \le t} V_s. \tag{9}$$

  MDD is a famous concept in hedge fund risk management, where drawdown denotes a decline from the past peak value M_t to the present value V_t.

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Shortly, this measure tells us the worst senario for a given investment horizon. That is, it represents how much loss an investor suffers from if he/she enter and exit an investment at the worst timing.

As it is widely recognized in practice that investment performance largely depends on its starting and exiting timing, MDD is thought to be an important measure. Namely, small MDD implies that an investor has not suffered from a large loss, whenever he/she starts the investment, at least on the past data.

- Sharpe ratio (ShR):

$$ShR \equiv (\bar{R} - r_f)/SD, \tag{10}$$

where r_f denotes a risk-free rate. In investment performance evaluation, risk-adjusted returns are often regarded as the most important measures. Among them, ShR is the most famous one, which is also a basic quantity in the field of financial economics. In this paper, we assume $r_f = 0$ because Bank of Japan guides short-term rates at -0.1% and the 10-year bond yield around 0% during most of the test period.

- Sortino ratio (SoR):

$$SoR \equiv (\bar{R} - r_f)/DD. \tag{11}$$

SoR is also useful because it adjusts risk by using DD, which makes it possible to focus on only downside risk.

- Sterling Ratio (StR):

$$StR \equiv (\bar{R} - r_f)/MDD. \tag{12}$$

StR is a measure of risk-adjusted return that uses drawdown measures as denominator.

References

[1] Yoo, P. D., Kim, M. H., & Jan, T. (2005, November). Machine learning techniques and use of event information for stock market prediction: A survey and evaluation. In International Conference on Computational Intelligence for Modelling, Control and Automation and International Conference on Intelligent Agents, Web Technologies and Internet Commerce (CIMCA-IAWTIC'06) (Vol. 2, pp. 835-841). IEEE.

[2] Fama, E. F. (1965). The behavior of stock-market prices. The journal of Business, 38(1), 34-105.

[3] Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. The journal of Finance, 25(2), 383-417.

[4] Campbell, J. Y., & Thompson, S. B. (2007). Predicting excess stock returns out of sample: Can anything beat the historical average?. The Review of Financial Studies, 21(4), 1509-1531.

[5] Nakano, M., Takahashi, A., & Takahashi, S. (2017). Generalized exponential moving average (EMA) model with particle filtering and anomaly detection. Expert Systems with Applications, 73, 187-200.

[6] Bengio, Y., Ducharme, R., & Vincent, P. (2000). A neural probabilistic language model. Advances in neural information processing systems, 13.

Electronic copy available at: https://ssrn.com/abstract=4511658

[7] Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. Journal of machine Learning research, 3(Jan), 993-1022.

[8] Pennington, J., Socher, R., & Manning, C. D. (2014, October). Glove: Global vectors for word representation. In Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP) (pp. 1532-1543).

[9] Lodhi, H., Saunders, C., Shawe-Taylor, J., Cristianini, N., & Watkins, C. (2002). Text classification using string kernels. Journal of machine learning research, 2(Feb), 419-444.

[10] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. Advances in neural information processing systems, 26.

[11] McCallum, A., & Nigam, K. (1998, July). A comparison of event models for naive bayes text classification. In AAAI-98 workshop on learning for text categorization (Vol. 752, No. 1, pp. 41-48).

[12] Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. The Journal of finance, 62(3), 1139-1168.

[13] Garcia, D. (2013). Sentiment during recessions. The journal of finance, 68(3), 1267-1300.

[14] Pagolu, V. S., Reddy, K. N., Panda, G., & Majhi, B. (2016, October). Sentiment analysis of Twitter data for predicting stock market movements. In 2016 international conference on signal processing, communication, power and embedded system (SCOPES) (pp. 1345-1350). IEEE.

[15] Sun, T., Wang, J., Zhang, P., Cao, Y., Liu, B., & Wang, D. (2017, August). Predicting stock price returns using microblog sentiment for chinese stock market. In 2017 3rd International Conference on Big Data Computing and Communications (BIGCOM) (pp. 87-96). IEEE.

[16] Hinton, G. E., Osindero, S., & Teh, Y. W. (2006). A fast learning algorithm for deep belief nets. Neural computation, 18(7), 1527-1554.

[17] Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. Mathematics of control, signals and systems, 2(4), 303-314.

[18] Ballings, M., Van den Poel, D., Hespeels, N., & Gryp, R. (2015). Evaluating multiple classifiers for stock price direction prediction. Expert Systems with Applications, 42(20), 7046-7056.

[19] Chong, E., Han, C., & Park, F. C. (2017). Deep learning networks for stock market analysis and prediction: Methodology, data representations, and case studies. Expert Systems with Applications, 83, 187-205.

[20] Huck, N. (2010). Pairs trading and outranking: The multi-step-ahead forecasting case. European Journal of Operational Research, 207(3), 1702-1716.

[21] de Oliveira, F. A., Nobre, C. N., & Zarate, L. E. (2013). Applying Artificial Neural Networks to prediction of stock price and improvement of the directional prediction index-Case study of PETR4, Petrobras, Brazil. Expert Systems with Applications, 40(18), 7596-7606.

Electronic copy available at: https://ssrn.com/abstract=4511658

[22] Nakano, M., Takahashi, A., & Takahashi, S. (2018). Bitcoin technical trading with artificial neural network. Physica A: Statistical Mechanics and its Applications, 510, 587-609.

[23] Nakano, M., & Takahashi, A. (2020). A new investment method with AutoEncoder: Applications to crypto currencies. Expert Systems with Applications, 162, 113730.

[24] Lopez-Lira, A., & Tang, Y. (2023). Can ChatGPT forecast stock price movements? return predictability and large language models. arXiv preprint arXiv:2304.07619.

[25] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in neural information processing systems, 30.

[26] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. Advances in neural information processing systems, 33, 1877-1901.

[27] Okimoto, T., & Hirasawa, E. (2014). Stock market predictability using news indexes. Security Analysis Journal, 52(4), 67-75.

[28] Goshima, K., & Takahashi, H. (2016). Quantifying news tone to analyze the Tokyo Stock Exchange with deep learning. Security Analysis Journal, 54(3), 76-86.

[29] Katayama, D., & Tsuda, K. (2018). A method of measurement of the impact of Japanese news on stock market. Procedia computer science, 126, 1336-1343.

[30] Akita, R., Yoshihara, A., Matsubara, T., & Uehara, K. (2016, June). Deep learning for stock prediction using numerical and textual information. In 2016 IEEE : ACIS 15th International Conference on Computer and Information Science (ICIS) (pp. 1-6). IEEE.

[31] Kudo, T. (2005). Mecab: Yet another part-of-speech and morphological analyzer. http://taku910.github.io/mecab/.

[32] Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). Biometrika, 52(3 : 4), 591-611.

[33] Wilcoxon, F. (1945). Individual Comparisons by Ranking Methods. Biometrics, 1(6), 80-83.

Electronic copy available at: https://ssrn.com/abstract=4511658